

SYED MAOZ ARIF

Hardware Developer & Electronics Engineer

President @ IET On Campus CUI Abbottabad

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PROFESSIONAL SUMMARY

Hardware developer and electronics engineering student (4th semester) at COMSATS University with hands-on experience in multi-layer PCB design, embedded systems, and IoT. Builder of 9 real-world projects, holder of 17 certifications, and winner of 5 awards including a global Instructables contest. Active freelancer on Fiverr & Upwork serving international clients. President at IET On Campus and Cabinet Member of the Youth Parliament Pakistan.

EDUCATION

BSc Electrical and Electronics Engineering (EEE)

COMSATS University Islamabad, Abbottabad Campus

2024 – 2028 (Expected)

4th Semester

- Student ID: FA24-EEE-014 | Focus: Embedded Systems, PCB Design, Signal Integrity, IoT
- Altium Global Scholarship Awardee (March 2026)

FSc Pre-Engineering

Peace Group of School & College, Abbottabad

Completed 2024

Higher Secondary

- Core Subjects: Mathematics, Physics, Analytical Reasoning

LEADERSHIP & PROFESSIONAL EXPERIENCE

President

IET On Campus, CUI Abbottabad

July 2026 – Present

Leadership

- Guiding chapter operations and representing the student branch to foster engineering excellence
- Organizing technical events and professional development initiatives for 50+ members

Membership Head

IET On Campus, Abbottabad

May 2025 – July 2026

Leadership

- Led membership initiatives and organized technical workshops (Computer Vision & AI in UAVs)
- Commended for extraordinary leadership, team management, and problem-solving abilities

Cabinet Member

Youth Parliament Pakistan

2025 – 2027 Session

Governance

- Serving in a leadership capacity for youth advocacy and national policy discussions

Freelance Hardware & Mechanical Designer

Fiverr & Upwork

Ongoing

Professional

- Professional multi-layer PCB design, schematic capture, and 3D CAD modeling for international clients
- Project scoping, deadline management, and professional client communication

KEY PROJECTS

IncliSense Precision Digital Level

Arduino Nano | MPU6050 | IoT | 3D Printing

2026 | Runner-Up, Instructables Sensors Contest (Global)

- Precision leveling device with dual-axis spatial data, 11-LED interface, integrated buzzer, and PC telemetry
- Custom 3D-printed ABS enclosure; documented and published on Instructables

Industrial Edge AI Condition Monitoring & 3D Digital Twin

SP26

Edge AI | Condition Monitoring | 3D Digital Twin | IoT

- Complex Engineering Problem project; secured 1st Position in Software Category at COMSATS University

First FPGA Programming – iCEbreaker

Verilog | iCEbreaker FPGA | Digital Logic | iceStorm Toolchain

2026

- LED blinking via Verilog HDL; learned digital logic synthesis, pin assignment, and bitstream generation

6S UAV Battery Management System (BMS)

BQ76930 | XT60 | MOSFETs | UAV Architecture

2026

- Complete 6-cell BMS schematic with charge/discharge protection MOSFETs and hardware telemetry

Mixed-Signal RF Transceiver Board

ESP32 | RF TX/RX | 50Ω Impedance Matching

2026

- 50Ω impedance-matched RF traces, unified ground plane with 312 stitching vias (Faraday cage)
Smart Lighting System with Obstacle Detection 2026
Altium Designer | ATMEGA328PB-AU | VL53L0X ToF | TP4056 USB-C
 - Custom PCB with Time-of-Flight sensor, dual LED arrays, and USB-C battery charging circuit
- Omni-Sense Node: Mixed-Signal DAQ** 2026
ESP32 | SPI | PWM | RFID | MQ-Series Sensors
 - Centralized DAQ system with 15 heterogeneous peripherals; dual-rail power distribution network
- Wi-Fi Router UPS Power Bank** 2026
Hardware Design | BMS | DC-DC Conversion | 3S Li-ion
 - Custom DC UPS with 7-segment voltage display, rotary adjustment, and fused overcurrent protection
- Personal Portfolio Website** 2026
HTML5 | CSS3 | JavaScript | Firebase Firestore | UI/UX
 - Responsive glassmorphism design with Firebase social features, AI assistant mode, and GPU-optimized animations

AWARDS & HONORS

- **Runner-Up – Instructables Sensors Contest 2026 (Global)** — Bronze Medal for “IncliSense” project April 2026
- **1st Position – Best CEP Project Award** — COMSATS University & District Youth Office Abbottabad July 2026
- **1st Position – Best Circuit Design** — COMSATS University & IET Hardware Workshop Feb 2026
- **Altium Global Scholarship Awardee** — Selected for emerging engineering excellence March 2026
- **Best Project Award** — COMSATS University – Outstanding engineering design & innovation Feb 2026

CERTIFICATIONS (17)

- Altium Certified: Designer Essentials
- Altium Education PCB Basic Design
- Generative AI Mastermind (Outskill)
- Arduino & IoT – Cardiff Institute
- Unleashing Creativity with AI
- Practical Entrepreneurship
- 3D Printing Certificate
- Solar Solution for Pakistan
- IET On Campus Leadership
- IET On Campus Recognition
- IET Experience Certificate
- Student Week SP26 Certificate
- Kangaroo Test Achievement
- Best Project Award Certificate
- Inovatex Project 2025
- Best CEP Project Award – 1st Position
- Organizer – Project Exhibition 2026

TECHNICAL SKILLS

PCB & Hardware	Altium Designer, Altium 365, CircuitStudio, KiCad, EasyEDA, Proteus, LTspice, Multisim
Embedded & IoT	ESP32, STM32, Arduino, ATMEGA328PB, Verilog (FPGA), MQTT, Firebase, Blynk
3D CAD	Autodesk Inventor, SolidWorks, Fusion 360, AutoCAD, Tinkercad
Programming	Python, MATLAB, C/C++ (Embedded), Verilog, HTML5, CSS3, JavaScript
Documentation	LaTeX, Overleaf, draw.io, Microsoft Office Suite
AI Tools	Gemini, ChatGPT, Claude, Google AI Studio, Microsoft Copilot
Design & Media	Adobe Photoshop, Canva, Inkscape

INTERPERSONAL SKILLS

Public Speaking | Leadership & Team Management | Communication & Client Relations | Collaboration & Empathy | Critical Thinking & Problem Solving | Project Scoping & Time Management

INTERESTS

Cricket | Trekking | Exploring New Places